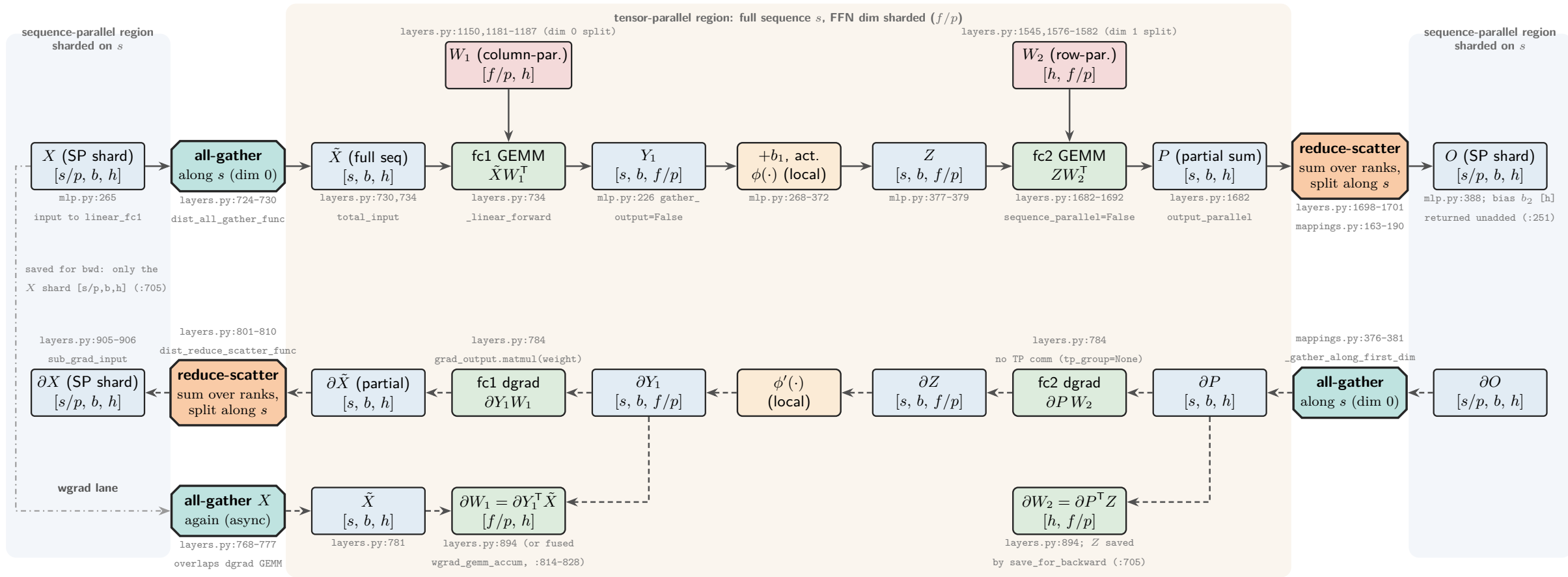


FORWARD ↑

↓ BACKWARD (dgrad)



Notation. s = sequence length, b = micro-batch, h = hidden_size, f = ffh_hidden_size, p = TP size; one rank shown. Weights stored as [out, in] (F .linear computes XW^T). With gated_linear_unit, fc1 output is doubled (mlp.py:207-208): W_1 $[2f/p, h]$, Y_1 $[s, b, 2f/p]$, and the GLU activation halves it back to Z $[s, b, f/p]$.

Comms. All TP collectives act on dim 0 (s). Forward (2): all-gather of X before fc1, reduce-scatter of P after fc2. Backward (3): all-gather of ∂O , reduce-scatter of $\partial \tilde{X}$, and a re-all-gather of X for ∂W_1 (async, overlapped with the fc1 dgrad GEMM). With SP, fc1 skips copy_to_tensor_model_parallel_region (layers.py:1340-1346) and allreduce_dgrad is False (layers.py:1256-1258), so the reduce-scatter replaces the all-reduce. Activation and b_1 are purely local.

Scope. Drawn from Megatron's native ColumnParallelLinear/RowParallelLinear + LinearWithGradAccumulationAndAsyncCommunication; Transformer-Engine-backed linear layers are not in the pinned files. Solid arrows: forward; dashed: backward; dash-dot gray: tensor saved for backward. Teal chamfered = all-gather, orange chamfered = reduce-scatter.